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Case Study

Case Study: Social Media Data Processing using NoSQL (MongoDB)

1. Objective

To process unstructured social media data using a NoSQL database (MongoDB) and perform analysis such as identifying trending hashtags.

2. Problem Statement

Social media platforms generate massive amounts of unstructured data (posts, hashtags, comments, likes). Traditional relational databases struggle with scalability and schema flexibility.

This case study focuses on:

- Storing social media posts in MongoDB
- Querying and analyzing data efficiently
- Extracting insights like trending hashtags

3. Tools & Technologies

- MongoDB (NoSQL Database)
- MongoDB Compass / Mongo Shell
- JSON / CSV dataset
- Optional: Python or Node.js for data import

4. Dataset Description

Sample dataset (posts.json):

```
[ {  
  "user": "john_doe",  
  "post": "Loving the new tech trends! #AI #MachineLearning",  
  "hashtags": ["AI", "MachineLearning"],
```

```
"likes": 120,  
"timestamp": "2026-04-01T10:00:00"  
}, {"user": "alice123",  
"post": "Workout done! #Fitness #Health",  
"hashtags": ["Fitness", "Health"],  
"likes": 80,  
"timestamp": "2026-04-02T08:30:00" },  
{"user": "tech_guru",  
"post": "Future is AI driven #AI #Innovation",  
"hashtags": ["AI", "Innovation"],  
"likes": 200,  
"timestamp": "2026-04-03T12:15:00"}]
```

5. Database Design (MongoDB)

- Database Name: socialMediaDB
- Collection Name: posts

MongoDB stores data in flexible JSON-like documents, making it ideal for unstructured social media content.

6. Implementation Steps

6.1 Start MongoDB

```
mongod
```

```
mongo
```

6.2 Create Database & Collection

```
use socialMediaDB
```

```
db.createCollection("posts")
```

6.3 Import Dataset

Using MongoDB import tool:

```
mongoimport --db socialMediaDB --collection posts --file posts.json --jsonArray
```

7. Queries for Data Processing

7.1 View All Posts

```
db.posts.find().pretty()
```

7.2 Filter Posts with Specific Hashtag (#AI)

```
db.posts.find({ hashtags: "AI" })
```

7.3 Find Posts with More Than 100 Likes

```
db.posts.find({ likes: { $gt: 100 } })
```

7.4 Find Top Trending Hashtags

```
db.posts.aggregate([  
  { $unwind: "$hashtags" },  
  { $group: { _id: "$hashtags", count: { $sum: 1 } } },  
  { $sort: { count: -1 } },  
  { $limit: 5 }  
])
```

7.5 Average Likes per Hashtag

```
db.posts.aggregate([  
  { $unwind: "$hashtags" },  
  { $group: { _id: "$hashtags", avgLikes: { $avg: "$likes" } } }  
])
```

7.6 Most Active Users

```
db.posts.aggregate([  
  { $group: { _id: "$user", totalPosts: { $sum: 1 } } },  
  { $sort: { totalPosts: -1 } }
```

])

8. Output Example

Top Trending Hashtags:

AI → 2

MachineLearning → 1

Fitness → 1

Health → 1

Innovation → 1

10. Applications

- **Social media analytics**
- **Trend detection**
- **Sentiment analysis systems**
- **Recommendation engines**

11. Conclusion

This case study demonstrates how MongoDB efficiently handles unstructured social media data and enables powerful analytics like identifying trending hashtags using aggregation queries.